

Continuous Kantorovich Formulation

Statements, hints and worked solutions

Each exercise is followed by question-by-question hints and a worked solution. Try the questions before reading the hints.

Exercise 1 (Couplings in basic cases). (a) Let α, β be finite *positive* measures with *different* total masses. What is $\Pi(\alpha, \beta)$, the set of couplings?

(b) (i) What are the couplings between δ_x and $\mathcal{U}[0, 1]$?

(ii) What are the couplings between $\frac{1}{2}(\delta_x + \delta_y)$ and $\mathcal{U}[0, 1]$, assuming $x \neq y$?

Useful hints

(a) Compare the total masses of the two marginals.

(b-i) The first coordinate is fixed.

(b-ii) Given $t \in [0, 1]$, let $q(t)$ be the conditional probability of the first coordinate being x ; impose its total mass.

Detailed solution

(a) Different total masses. Let $M_\alpha := \alpha(X)$ and $M_\beta := \beta(X)$. For any coupling π ,

$$\pi(X \times X) = \alpha(X) = M_\alpha = \beta(X) = \pi(X \times X),$$

so $M_\alpha = M_\beta$ is necessary. If $M_\alpha \neq M_\beta$ then *no* coupling exists:

$$\Pi(\alpha, \beta) = \emptyset.$$

(b-i) δ_x vs. $\mathcal{U}[0, 1]$. A coupling must live on $\{x\} \times [0, 1]$ and have second marginal $\mathcal{U}[0, 1]$. Disintegrating w.r.t. the first coordinate gives a unique conditional law on $[0, 1]$, namely $\mathcal{U}[0, 1]$. Hence the unique coupling is

$$\pi = \delta_x \otimes \mathcal{U}[0, 1].$$

(b-ii) $\frac{1}{2}(\delta_x + \delta_y)$ vs. $\mathcal{U}[0, 1]$. There are *many* couplings. Two equivalent descriptions:

(1) Disintegration w.r.t. the *second* marginal (on $[0, 1]$). Any coupling π with second marginal $\mathcal{U}[0, 1]$ can be written as

$$\pi(du, dt) = [q(t) \delta_x(du) + (1 - q(t)) \delta_y(du)] dt,$$

for some measurable $q : [0, 1] \rightarrow [0, 1]$ (here dt denotes Lebesgue/Uniform measure on $[0, 1]$). The first marginal is then

$$(\text{proj}_1)_\# \pi = \left(\int_0^1 q(t) dt \right) \delta_x + \left(1 - \int_0^1 q(t) dt \right) \delta_y.$$

Requiring $(\text{proj}_1)_\# \pi = \frac{1}{2}(\delta_x + \delta_y)$ is equivalent to the single constraint

$$\int_0^1 q(t) dt = \frac{1}{2}.$$

Thus all couplings are exactly the measures π_q in the displayed disintegration, indexed by measurable $0 \leq q \leq 1$ with $\int_0^1 q(t) dt = 1/2$.

(2) Disintegration w.r.t. the *first* marginal (on $\{x, y\}$). Equivalently, any such coupling can be written as

$$\pi = \frac{1}{2} \delta_x \otimes \mu_x + \frac{1}{2} \delta_y \otimes \mu_y,$$

where μ_x, μ_y are probabilities on $[0, 1]$ with

$$\frac{\mu_x + \mu_y}{2} = \mathcal{U}[0, 1] \iff \mu_x(B) + \mu_y(B) = 2|B| \text{ for all Borel } B \subset [0, 1].$$

The two parameterizations are linked by $\mu_x = 2q(t) dt$ and $\mu_y = 2(1 - q(t)) dt$. These conditional laws necessarily have densities because their average is Lebesgue measure.

Extreme examples. Take a measurable $A \subset [0, 1]$ with $|A| = \frac{1}{2}$ and set

$$q(t) = \mathbf{1}_A(t).$$

Then

$$\pi_A(du, dt) = [\mathbf{1}_A(t) \delta_x + (1 - \mathbf{1}_A(t)) \delta_y](du) dt = \frac{1}{2} \delta_x \otimes \mathcal{U}(A) + \frac{1}{2} \delta_y \otimes \mathcal{U}(A^c),$$

where $\mathcal{U}(A)$ and $\mathcal{U}(A^c)$ are the uniform laws on A and A^c (normalized). These “split-by-partition” couplings are extremal points of the convex set above.

Exercise 2 (Gaussian couplings in $\mathbb{R}$ (zero means)).

Let $\alpha = \mathcal{N}(0, \sigma_\alpha^2)$ and $\beta = \mathcal{N}(0, \sigma_\beta^2)$ with $\sigma_\alpha, \sigma_\beta \geq 0$. Consider *Gaussian* couplings $\pi = \mathcal{N}(0, \Sigma)$ on $\mathbb{R} \times \mathbb{R}$ with

$$\Sigma = \begin{pmatrix} \sigma_\alpha^2 & c \\ c & \sigma_\beta^2 \end{pmatrix} \geq 0.$$

- Express the quadratic OT cost $\mathbb{E}[(X - Y)^2]$ in terms of c .
- Determine the constraint on c implied by $\Sigma \geq 0$.
- Solve the Kantorovich problem *restricted to Gaussian couplings* by minimizing over feasible c .
- What is the rank of Σ as a function of $(\sigma_\alpha, \sigma_\beta, c)$, and what is the support of π in each case? Compare the optimal Gaussian coupling with the conclusion of Brenier’s theorem.

Useful hints

- Expand $\mathbb{E}(X - Y)^2$.
- A symmetric 2×2 matrix is PSD exactly when both diagonal entries and the determinant are nonnegative.
- Maximize the covariance c . Cauchy–Schwarz gives the same bound for non-Gaussian couplings.
- A Gaussian is supported on the range of its covariance matrix.

Detailed solution

(a) Cost as a function of c .

$$\mathbb{E}[(X - Y)^2] = \mathbb{E}[X^2] + \mathbb{E}[Y^2] - 2\mathbb{E}[XY] = \sigma_\alpha^2 + \sigma_\beta^2 - 2c.$$

(b) Feasible range of c from $\Sigma \geq 0$. For a 2×2 symmetric $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$, PSD $\Leftrightarrow a \geq 0, b \geq 0, ab - c^2 \geq 0$. Thus

$$-\sigma_\alpha \sigma_\beta \leq c \leq \sigma_\alpha \sigma_\beta.$$

(c) **Minimization over feasible c .** The objective $\sigma_\alpha^2 + \sigma_\beta^2 - 2c$ decreases with c , so the minimum is at

$$c^* = \sigma_\alpha \sigma_\beta,$$

yielding

$$\mathcal{W}_2^2(\alpha, \beta) = \sigma_\alpha^2 + \sigma_\beta^2 - 2\sigma_\alpha \sigma_\beta = (\sigma_\alpha - \sigma_\beta)^2.$$

For any coupling, Cauchy–Schwarz gives $\mathbb{E}[XY] \leq \sigma_\alpha \sigma_\beta$, so the Gaussian optimizer is also globally optimal. At c^* (with $\sigma_\alpha > 0$), the optimal coupling is concentrated on $y = \frac{\sigma_\beta}{\sigma_\alpha}x$ (Monge map $T(x) = \frac{\sigma_\beta}{\sigma_\alpha}x$). If $\sigma_\alpha = 0$, feasibility forces $c = 0$ and $\mathcal{W}_2^2 = \sigma_\beta^2$.

(d) **Rank of Σ , support of π , and Brenier.**

- *Generic feasible c with $\sigma_\alpha, \sigma_\beta > 0$.* $\det \Sigma = \sigma_\alpha^2 \sigma_\beta^2 - c^2$. If $|c| < \sigma_\alpha \sigma_\beta$, then $\det \Sigma > 0$ and $\text{rank}(\Sigma) = 2$; π is a non-degenerate Gaussian with a density on $\mathbb{R}^2$ (full support).
- *Extremal $c = \pm \sigma_\alpha \sigma_\beta$ with $\sigma_\alpha, \sigma_\beta > 0$.* Then $\det \Sigma = 0$ and $\text{rank}(\Sigma) = 1$. For $c = +\sigma_\alpha \sigma_\beta$ one has correlation +1 and

$$Y = \frac{\sigma_\beta}{\sigma_\alpha} X \quad \text{a.s.,}$$

so π is supported on the line $y = \frac{\sigma_\beta}{\sigma_\alpha}x$. (For $c = -\sigma_\alpha \sigma_\beta$, the line is $y = -\frac{\sigma_\beta}{\sigma_\alpha}x$.)

- *Singular marginals.* If $\sigma_\alpha = 0$ and $\sigma_\beta > 0$, PSD enforces $c = 0$ and $\Sigma = \begin{pmatrix} 0 & 0 \\ 0 & \sigma_\beta^2 \end{pmatrix}$ has rank 1; $X = 0$ a.s. and $Y \sim \mathcal{N}(0, \sigma_\beta^2)$, so π lives on the vertical line $\{x = 0\}$. The case $\sigma_\beta = 0$ is symmetric (horizontal line). If $\sigma_\alpha = \sigma_\beta = 0$, then $\Sigma = 0$ and $\text{rank}(\Sigma) = 0$ (a single atom at $(0, 0)$).
- *Comparison with Brenier.* When $\sigma_\alpha > 0$ (so α is absolutely continuous), Brenier's theorem ensures the global $\mathcal{W}_2$ -optimal coupling is induced by a transport map $T = \nabla \phi$. In 1D this map is increasing; for Gaussians it is linear: $T(x) = \frac{\sigma_\beta}{\sigma_\alpha}x$. This agrees with the Gaussian-restricted optimizer above: at $c = c^*$ the coupling has rank 1 and is supported on the graph of T . When α is not absolutely continuous (e.g. $\sigma_\alpha = 0$), Brenier's conclusion need not hold; indeed the optimal plan cannot be represented by a deterministic map from α to a non-degenerate β .

Exercise 3 (Uniform $\rightarrow$ two Diracs; two Diracs $\rightarrow$ uniform).

Fix two distinct points $x \neq y$ in $\mathbb{R}$ and let

$$\alpha = \mathcal{U}[0, 1], \quad \beta = \frac{1}{2} \delta_x + \frac{1}{2} \delta_y.$$

Consider the quadratic cost $c(u, v) = |u - v|^2$ (the increasing rearrangement remains optimal for $|u - v|^p$, $p \geq 1$, but uniqueness may fail for $p = 1$).

- Find the $\mathcal{W}_p$ -optimal Monge map $T : [0, 1] \rightarrow \mathbb{R}$ that pushes α to β .
- Is there a Monge map $S : \mathbb{R} \rightarrow [0, 1]$ such that $S_\# \beta = \alpha$?
- Describe an optimal Kantorovich coupling $\pi \in \Pi(\alpha, \beta)$ between α and β .

Useful hints

- Order the two target atoms and match quantiles.
- A map sends two atoms to at most two atoms.
- Use the graph plan of the map in (a), and transpose it for the reverse direction.

Detailed solution

Setup. Without loss of generality, reorder the atoms so that $x < y$ (if $y < x$, swap the names x and y below).

(a) Optimal Monge map $\alpha \rightarrow \beta$. In 1D and for convex costs, the optimal map is the increasing rearrangement (quantile map)

$$T = F_\beta^{-1} \circ F_\alpha.$$

Here $F_\alpha(t) = t$ on $[0, 1]$ and $F_\beta^{-1}(u) = \begin{cases} x, & u \in (0, \frac{1}{2}], \\ y, & u \in (\frac{1}{2}, 1]. \end{cases}$ Hence

$$T(t) = \begin{cases} x, & t \in (0, \frac{1}{2}], \\ y, & t \in (\frac{1}{2}, 1]. \end{cases}$$

This T pushes α to β and is optimal among maps; for the stated quadratic cost it is unique up to null sets, since exchanging a pair assigned in reversed order strictly lowers the cost. For $p = 1$, other partitions can be optimal.

(b) No Monge map $\beta \rightarrow \alpha$. If S were a measurable map with $S_\# \beta = \alpha$, then

$$S_\# \beta = \frac{1}{2} \delta_{S(x)} + \frac{1}{2} \delta_{S(y)},$$

which is purely atomic (at most two atoms) and cannot equal the continuous law $\mathcal{U}[0, 1]$. Therefore no Monge map exists from β to α .

(c) An optimal coupling π . Push forward α by the optimal map T and keep track of preimages:

$$\pi = (\text{id}, T)_\# \alpha = (\lambda|_{[0, 1/2]}) \otimes \delta_x + (\lambda|_{(1/2, 1]}) \otimes \delta_y,$$

where λ is Lebesgue on $[0, 1]$. Equivalently, for Borel $A \subset [0, 1]$, $B \subset \mathbb{R}$,

$$\pi(A \times B) = \int_{A \cap [0, 1/2]} \mathbf{1}_B(x) dt + \int_{A \cap (1/2, 1]} \mathbf{1}_B(y) dt.$$

This coupling has first marginal α and second marginal β , and it is optimal (it is induced by the optimal map).

Optional cost value (for $p = 2$).

$$W_2^2(\alpha, \beta) = \int_0^{1/2} (t - x)^2 dt + \int_{1/2}^1 (t - y)^2 dt.$$

Exercise 4 (Bottleneck cost: metric property and closed-form OT value).

Let $(X, \mathcal{B})$ be a standard Borel space (so the diagonal in $X \times X$ is measurable) and let $m : X \rightarrow (0, \infty)$ be measurable (strictly positive everywhere). Define

$$c(x, y) := \begin{cases} 0, & x = y, \\ m(x) + m(y), & x \neq y. \end{cases}$$

(a) Show that $d(x, y) := c(x, y)$ is a metric on X .

(b) For probability measures α, β on $(X, \mathcal{B})$ with $\int m d\alpha + \int m d\beta < \infty$, compute the Kantorovich value

$$W_c(\alpha, \beta) := \inf_{\pi \in \Pi(\alpha, \beta)} \int c(x, y) d\pi(x, y)$$

in closed form.

(c) Give an explicit optimal plan in terms of the maximal common submeasure $\mu := \alpha \wedge \beta$.

Useful hints

- (a) Separate the case of coincident points from three distinct points.
- (b) How much mass can remain on the diagonal at zero cost?
- (c) Couple the two mutually singular residual measures by a normalized product, with a separate convention if their mass is zero.

Detailed solution

(a) $d = c$ is a metric. Nonnegativity and symmetry are immediate. Also $d(x, y) = 0$ iff $x = y$: if $x \neq y$, then $d(x, y) = m(x) + m(y) > 0$ by the strict positivity of m .

For the triangle inequality, fix $x, y, z \in X$. If any two coincide, the claim is obvious. When x, y, z are pairwise distinct,

$$d(x, z) = m(x) + m(z) \leq (m(x) + m(y)) + (m(y) + m(z)) = d(x, y) + d(y, z),$$

since $m(y) > 0$. Hence d is a metric.

(b) Closed form for $W_c(\alpha, \beta)$. Let $\mu := \alpha \wedge \beta$ be the greatest measure dominated by both α and β . Recall the Hahn decomposition $\alpha - \beta = (\alpha - \mu) - (\beta - \mu)$ and

$$|\alpha - \beta| = (\alpha - \mu) + (\beta - \mu).$$

Upper bound (constructive). Keep the common mass on the diagonal and match only the residuals off the diagonal. Formally, choose any $\gamma \in \Pi(\alpha - \mu, \beta - \mu)$ with support in $\{(x, y) : x \neq y\}$ and set

$$\pi^* := \mu \circ (x \mapsto (x, x))^{-1} + \gamma \in \Pi(\alpha, \beta).$$

Then

$$\begin{aligned} \int c \, d\pi^* &= \int (m(x) + m(y)) \, d\gamma(x, y) \\ &= \int m \, d(\alpha - \mu) + \int m \, d(\beta - \mu) = \int m \, d|\alpha - \beta|. \end{aligned}$$

Therefore $W_c(\alpha, \beta) \leq \int m \, d|\alpha - \beta|$.

Lower bound (decomposition). For any $\pi \in \Pi(\alpha, \beta)$, decompose $\pi = \pi_{\text{diag}} + \pi_{\text{off}}$ with π_{diag} supported on $\{x = y\}$ and π_{off} on $\{x \neq y\}$. The diagonal part costs 0. Denote the marginals of π_{off} by $\alpha_{\text{off}}, \beta_{\text{off}}$. Since at most μ -mass can stay on the diagonal, one has $\alpha_{\text{off}} \geq \alpha - \mu$ and $\beta_{\text{off}} \geq \beta - \mu$. Thus

$$\begin{aligned} \int c \, d\pi &= \int m \, d\alpha_{\text{off}} + \int m \, d\beta_{\text{off}} \\ &\geq \int m \, d(\alpha - \mu) + \int m \, d(\beta - \mu) = \int m \, d|\alpha - \beta|. \end{aligned}$$

Taking the infimum over π yields $W_c(\alpha, \beta) \geq \int m \, d|\alpha - \beta|$.

Combining bounds,

$$W_c(\alpha, \beta) = \int_X m \, d|\alpha - \beta|.$$

Dual check. Choose a measurable representative of $h = d(\alpha - \beta)/d|\alpha - \beta|$ with $|h| \leq 1$ everywhere, and put $\varphi = mh$, $\psi = -\varphi$. Then $|\varphi(x)| \leq m(x)$, so $\varphi(x) - \varphi(y) \leq c(x, y)$ for $x \neq y$, and it is zero on the diagonal. This pair attains $\int m \, d|\alpha - \beta|$.

(c) An explicit optimal coupling. With $\mu = \alpha \wedge \beta$ as above, *any* choice of $\gamma \in \Pi(\alpha - \mu, \beta - \mu)$ (explicitly, $\gamma = (\alpha - \mu) \otimes (\beta - \mu)/r$ if $r := 1 - \mu(X) > 0$, and $\gamma = 0$ if $r = 0$) and

$$\pi^\star = \mu \circ (x \mapsto (x, x))^{-1} + \gamma$$

is optimal and attains the value $\int m d|\alpha - \beta|$. All common mass stays on the diagonal (zero cost), while every unit of unmatched mass pays $m(x)$ at departure and $m(y)$ at arrival, independently of how the residuals are paired.

Further exercises

Exercise 5 (Gaussian couplings in $\mathbb{R}^d$ (zero means)).

Let $\alpha = \mathcal{N}(0, \Sigma_\alpha)$ and $\beta = \mathcal{N}(0, \Sigma_\beta)$ with $\Sigma_\alpha, \Sigma_\beta \in \mathbb{S}_{++}^d$. Any Gaussian coupling on $\mathbb{R}^d \times \mathbb{R}^d$ has block covariance

$$\Sigma = \begin{pmatrix} \Sigma_\alpha & K \\ K^\top & \Sigma_\beta \end{pmatrix} \geq 0.$$

- Express the quadratic OT cost $\mathbb{E}\|X - Y\|^2$ in terms of K .
- Determine the constraint on K implied by $\Sigma \geq 0$ (give equivalent forms).
- Solve the Kantorovich problem *restricted to Gaussian couplings* by minimizing the cost over feasible K .
- What is the rank of Σ and the support of the coupling π for (i) strict feasibility and (ii) an optimizer from part (c)? Compare the optimal Gaussian coupling with Brenier's theorem.

Useful hints

- Expand the squared norm and take traces.
- Take the Schur complement of Σ_α .
- Whiten K and maximize a trace over contractions; be careful about the orientation of the SVD.
- Inspect the Schur complement and the graph of the linear optimal map.

Detailed solution

(a) Cost as a function of K .

$$\mathbb{E}\|X - Y\|^2 = \mathbb{E}\|X\|^2 + \mathbb{E}\|Y\|^2 - 2\mathbb{E}\langle X, Y \rangle = \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta) - 2\text{tr}(K).$$

(b) Feasible K from $\Sigma \geq 0$. Since $\Sigma_\alpha > 0$, Schur complement gives

$$\Sigma \geq 0 \iff \Sigma_\beta - K^\top \Sigma_\alpha^{-1} K \geq 0.$$

Equivalently, with $M := \Sigma_\alpha^{-1/2} K \Sigma_\beta^{-1/2}$, this is $I - M^\top M \geq 0$, i.e. $\|M\|_{\text{op}} \leq 1$. Thus the feasible K are exactly

$$K = \Sigma_\alpha^{1/2} M \Sigma_\beta^{1/2} \quad \text{with} \quad \|M\|_{\text{op}} \leq 1.$$

Another equivalent form is $K \Sigma_\beta^{-1} K^\top \leq \Sigma_\alpha$.

(c) Minimization over feasible K . Using the parametrization in (b),

$$\text{tr}(K) = \text{tr}(M S), \quad S := \Sigma_\beta^{1/2} \Sigma_\alpha^{1/2}.$$

By von Neumann's trace inequality,

$$\max_{\|M\|_{\text{op}} \leq 1} \text{tr}(MS) = \text{tr}((S^\top S)^{1/2}) = \text{tr}\left((\Sigma_\beta^{1/2} \Sigma_\alpha \Sigma_\beta^{1/2})^{1/2}\right),$$

achieved for $M = VU^\top$ when $S = U \text{diag}(s) V^\top$ is an SVD. Therefore an optimal cross-covariance is

$$K^* = \Sigma_\alpha^{1/2} (\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2} \Sigma_\alpha^{-1/2}.$$

Every coupling, Gaussian or not, has a PSD block covariance and obeys the same bound. Since a Gaussian realizes the maximizing cross-covariance, plugging into (a) yields the global Kantorovich value

$$\mathcal{W}_2^2(\alpha, \beta) = \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta) - 2 \text{tr}\left((\Sigma_\beta^{1/2} \Sigma_\alpha \Sigma_\beta^{1/2})^{1/2}\right).$$

This is attained by the deterministic linear map

$$T(x) = Ax, \quad A = \Sigma_\alpha^{-1/2} (\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2} \Sigma_\alpha^{-1/2},$$

for which $\text{Cov}(X, T(X)) = \Sigma_\alpha A^\top = \Sigma_\alpha A = K^*$.

(d) Rank of Σ , support of π , and Brenier. Write the joint covariance as

$$\Sigma = \begin{pmatrix} \Sigma_\alpha & K \\ K^\top & \Sigma_\beta \end{pmatrix}.$$

- *Strict feasibility:* if $\Sigma_\beta - K^\top \Sigma_\alpha^{-1} K > 0$ (equivalently $\|M\|_{\text{op}} < 1$), then $\Sigma > 0$ and $\text{rank}(\Sigma) = 2d$. The Gaussian π has a smooth density on $\mathbb{R}^{2d}$ (full support).
- *At the optimizer $K = K^*$:* the Schur complement vanishes,

$$\Sigma_\beta - (K^*)^\top \Sigma_\alpha^{-1} K^* = 0,$$

so $\text{rank}(\Sigma) = d$. In fact (X, Y) satisfies $Y = AX$ a.s., hence

$$\text{supp}(\pi) = \{(x, Ax) : x \in \mathbb{R}^d\},$$

a d -dimensional linear subspace (the graph of T).

- *Brenier comparison:* since α is absolutely continuous, Brenier's theorem asserts that the *globally* $\mathcal{W}_2$ -optimal coupling is induced by a map $T = \nabla \phi$. For Gaussians, T is linear, symmetric positive semidefinite, and equals the A above. Thus the Gaussian-restricted optimizer coincides with the Brenier map, and the joint law is concentrated on its graph.

Exercise 6 (A non-Monge optimal plan in $\mathbb{R}^2$).

Let α be the uniform probability measure on the vertical segment

$$S_0 := \{(0, s) : s \in [0, 1]\}$$

(with respect to 1D length, normalized), and let

$$\beta := \frac{1}{2} \mathcal{U}(S_-) + \frac{1}{2} \mathcal{U}(S_+), \quad S_\pm := \{(\pm 1, t) : t \in [0, 1]\}.$$

We consider the quadratic cost $c(x, y) = \|x - y\|^2$ on $\mathbb{R}^2 \times \mathbb{R}^2$.

(a) **(Lower bound)** Show that for any coupling $\pi \in \Pi(\alpha, \beta)$,

$$\int \|x - y\|^2 d\pi(x, y) \geq 1.$$

(b) **(An optimal coupling)** Define $\pi^\star$ by the disintegration: for $X = (0, s) \sim \alpha$,

$$\pi^\star(\cdot \mid X = (0, s)) = \frac{1}{2} \delta_{(-1, s)} + \frac{1}{2} \delta_{(1, s)}.$$

Check that $\pi^\star \in \Pi(\alpha, \beta)$ and compute its cost. Conclude that the Kantorovich value equals 1.

(c) **(No Monge minimizer)** Let $T : S_0 \rightarrow S_- \cup S_+$ be a measurable map with $T_\# \alpha = \beta$. Show that

$$\int \|x - T(x)\|^2 d\alpha(x) > 1,$$

hence no Monge map attains the Kantorovich minimum.

(d) **(Structure of the optimizer and comparison with Brenier)** Describe $\text{supp}(\pi^\star)$ and explain why it is not contained in the graph of any map $x \mapsto T(x)$. Comment on why this does not contradict Brenier's theorem.

Useful hints

- (a) The horizontal displacement always has squared length 1.
- (b) Verify each target branch receives half of Lebesgue measure.
- (c) Cost 1 forces the vertical coordinate to remain unchanged. Can a measurable indicator equal $1/2$ almost everywhere?
- (d) Distinguish absolute continuity on a segment from absolute continuity in the ambient plane.

Detailed solution

(a) Lower bound. Write $x = (0, s)$ and $y = (\xi, t)$ with $\xi \in \{-1, +1\}$ and $s, t \in [0, 1]$. Then

$$\|x - y\|^2 = (\xi - 0)^2 + (t - s)^2 = 1 + (t - s)^2 \geq 1.$$

Integrating over any coupling π gives $\int \|x - y\|^2 d\pi \geq 1$.

(b) Optimal coupling $\pi^\star$. By construction, for any Borel $J \subset [0, 1]$,

$$\pi^\star(S_0 \times (\{-1\} \times J)) = \int_{s \in J} \frac{1}{2} ds = \frac{1}{2} |J|,$$

and similarly for S_+ . Hence $(\text{proj}_Y)_\# \pi^\star = \beta$, and trivially $(\text{proj}_X)_\# \pi^\star = \alpha$. Moreover, under $\pi^\star$ we have $t = s$ a.s., so

$$\int \|x - y\|^2 d\pi^\star = \int (1 + (t - s)^2) d\pi^\star = \int 1 d\pi^\star = 1.$$

Combined with (a), this proves that $\pi^\star$ is optimal and the Kantorovich value equals 1.

(c) No Monge minimizer. Suppose $T_\# \alpha = \beta$ with $T(0, s) = (\xi(s), u(s))$, where $\xi(s) \in \{-1, +1\}$ and $u(s) \in [0, 1]$. Because β assigns mass $\frac{1}{2}$ to each vertical segment with *uniform* t -coordinate, we must have

$$\lambda(\{s : \xi(s) = -1\}) = \lambda(\{s : \xi(s) = +1\}) = \frac{1}{2},$$

and the pushforward of Lebesgue restricted to $A_- := \{s : \xi(s) = -1\}$ by u must equal $\frac{1}{2} \mathbf{1}_{[0, 1]} dt$ (and likewise for A_+). Now compute the cost:

$$\int \|x - T(x)\|^2 d\alpha(x) = \int_0^1 (1 + (u(s) - s)^2) ds = 1 + \int_0^1 (u(s) - s)^2 ds.$$

If this equalled 1, then $(u(s) - s)^2 = 0$ a.e., hence $u(s) = s$ a.e. But then the t -marginal on the left segment is the image of Lebesgue by $s \mapsto s$ restricted to A_- , i.e. Lebesgue restricted to A_- . Equality with $\frac{1}{2}\mathbf{1}_{[0,1]} dt$ would require $\mathbf{1}_{A_-} = 1/2$ almost everywhere, which is impossible. Therefore $\int_0^1 (u(s) - s)^2 ds > 0$, and so every Monge map has cost > 1 . Consequently, no Monge map is optimal.

(d) Support and Brenier. The optimizer π^* is supported on two graphs:

$$\text{supp}(\pi^*) = \{((0, s), (-1, s)) : s \in [0, 1]\} \cup \{((0, s), (+1, s)) : s \in [0, 1]\}.$$

It is not contained in the graph of any single-valued map T , since for every $x = (0, s)$ the conditional law assigns positive mass to two distinct points $(-1, s)$ and $(+1, s)$. This does not contradict Brenier's theorem: that theorem requires the source to be absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^2$. Here α is supported on a 1D set (a segment), so the assumption fails and an optimal plan need not be induced by a map.

Remark (approximating the optimum with maps). Although no map attains cost 1, one can build maps with cost arbitrarily close to 1 by partitioning $[0, 1]$ into many small pieces, sending half of them to S_- and half to S_+ , and rearranging $u(s)$ inside each piece to match the uniform t -marginal; the added vertical cost can be made as small as $O(n^{-2})$ with n pieces.